

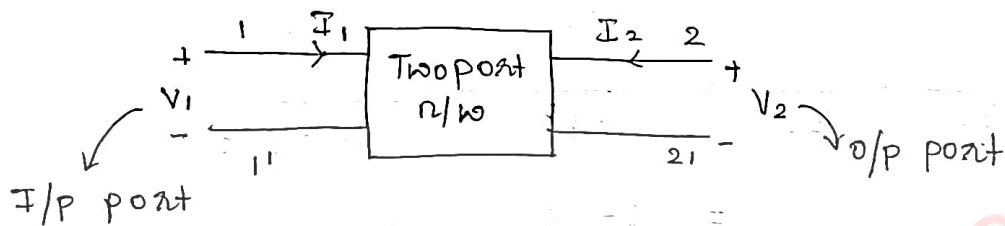
## Module - 5

### Two Port network parameters

#### Port

A port is a pair of terminals across which energy source (voltage or current source) is applied to the network.

Two port network consisting of two pairs of terminals one pair is for i/p side & another pair is for o/p side



Parameters.

dependent variable Independent variables

|                 |            |            |
|-----------------|------------|------------|
| (1) Z-parameter | $V_1, V_2$ | $I_1, I_2$ |
| (2) Y-parameter | $I_1, I_2$ | $V_1, V_2$ |
| (3) h-parameter | $V_1, I_2$ | $I_1, V_2$ |
| (4) T-parameter | $V_1, I_1$ | $V_2, I_2$ |

(1) Z-parameter:

$$V_1 = f(I_1, I_2)$$

$$V_2 = f(I_1, I_2)$$

The defining eqn for z-parameters are

$$V_1 = z_{11} I_1 + z_{12} I_2 \quad \text{--- (1)}$$

$$V_2 = z_{21} I_1 + z_{22} I_2 \quad \text{--- (2)}$$

To find  $z_{11}$  &  $z_{21}$ , make  $I_2 = 0$  (open ckt the o/p port)

$$z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0}, \quad z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0}.$$

To find  $Z_{12}$  &  $Z_{22}$ , make  $I_1 = 0$  [open ckt the i/p port]

$$Z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0} \quad Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0}$$

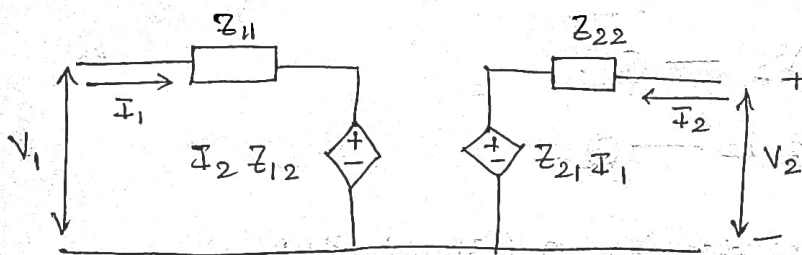
where  $Z_{11}$  = i/p impedance with o/p port open ckt

$Z_{22}$  = o/p impedance with i/p port open ckt

$Z_{12}$  = Transfer impedance with i/p port open ckt.

$Z_{21}$  = Transfer impedance with o/p port open ckt.

equivalent ckt



(2) Y parameters [short ckt admittance parameters];

$$I_1 = f(V_1, V_2)$$

$$I_2 = f(V_1, V_2)$$

The defining eqn for Y-parameters are

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \text{--- (2)}$$

To find  $Y_{11}$  &  $Y_{21}$  make  $V_2 = 0$  (short ckt o/p port)

$$Y_{11} = \frac{I_1}{V_1} \Big|_{V_2=0} \quad Y_{21} = \frac{I_2}{V_1} \Big|_{V_2=0}$$

To find  $Y_{12}$  &  $Y_{22}$  make  $V_1 = 0$  (short ckt i/p port)

$$Y_{12} = \frac{I_1}{V_2} \Big|_{V_1=0} \quad Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0}$$

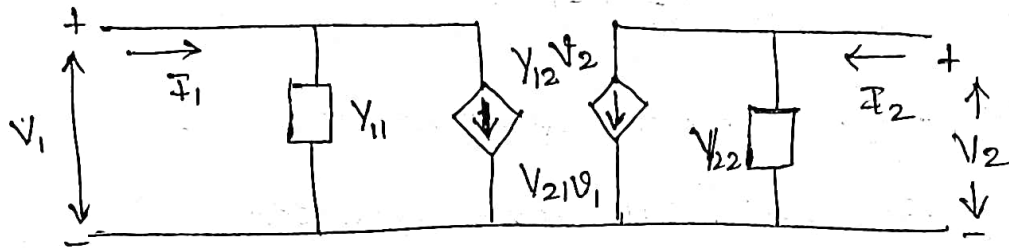
where,  $Y_{11}$  = i/p admittance with o/p port short ckt

$Y_{22}$  = o/p admittance with i/p port short ckt

$Y_{12}$  = Transfer admittance with i/p port short ckt.

$Y_{21}$  = Transfer admittance with o/p port short ckt.

equivalent ckt:-



(3) H - parameters :-

$$V_1 = f(I_1, V_2), \quad I_2 = f(I_1, V_2)$$

Defining eqn of H - parameters are.

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

To find  $h_{11}$  &  $h_{21}$  make  $V_2 = 0$  (short ckt)

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0}$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0}$$

To find  $h_{12}$  &  $h_{22}$  make  $I_1 = 0$  (open ckt)

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0}$$

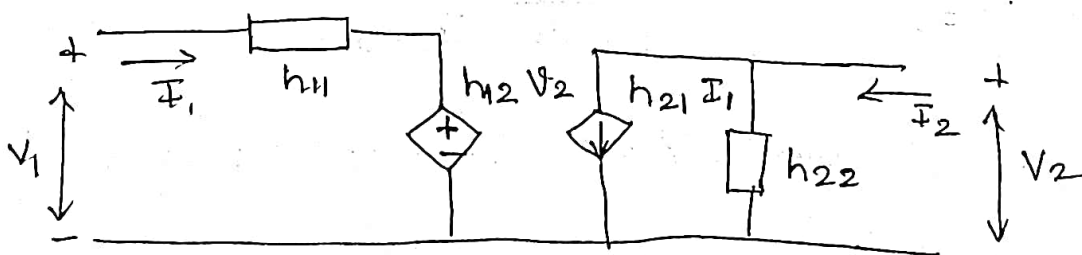
$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$

where  $h_{11}$  = i/p impedance with o/p port short ckt

$h_{21}$  = forward current gain with o/p port short ckt

$h_{12}$  = Reverse voltage gain with o/p port open ckt

$h_{22}$  = o/p admittance with i/p port open ckt.



(A) T-parameters (or) ABCD-parameters.

$$V_1 = f(V_2, I_2) \quad I_1 = f(V_2, I_2).$$

The defining eqn of T-parameters are

$$V_1 = A V_2 - B I_2 \quad \text{--- ①}$$

$$I_1 = C V_2 - D I_2 \quad \text{--- ②}$$

To find A & C make  $I_2 = 0$  (open ckt the o/p port)

$$A = \left. \frac{V_1}{V_2} \right|_{I_2=0} \quad C = \left. \frac{I_1}{V_2} \right|_{I_2=0}$$

To find B & D make  $V_2 = 0$ .

$$B = - \left. \frac{V_1}{I_2} \right|_{V_2=0} \quad D = - \left. \frac{I_1}{I_2} \right|_{V_2=0}$$

\* Relation b/w Z & Y-parameters.

Z-parameters in terms of Y-parameters:

The defining eqn for Z-parameters

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

In matrix form

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

The defining eqn of Y-parameters are

$$I_1 = Y_{11} V_1 + Y_{12} V_2, \quad I_2 = Y_{21} V_1 + Y_{22} V_2$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$[I] = [Y] [V]$$

$$\text{W.K.T } [Z] = \frac{1}{[Y]} \Rightarrow Z = [Y]^{-1}$$



$$Z = \frac{\text{adj } Y}{\Delta Y}$$

$$\text{where } Y = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \frac{\begin{bmatrix} Y_{22} & -Y_{12} \\ -Y_{21} & Y_{11} \end{bmatrix}}{\Delta Y} = \begin{bmatrix} Y_{22}/\Delta Y & -Y_{12}/\Delta Y \\ -Y_{21}/\Delta Y & Y_{11}/\Delta Y \end{bmatrix}$$

$$\text{where } \Delta Y = Y_{11} Y_{22} - Y_{21} Y_{12}$$

\* Y-parameters in terms of Z-parameters

$$Y = \frac{1}{[Z]} = [Z]^{-1} = \frac{\text{adj } Z}{\Delta Z}$$

$$\text{where } [Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \frac{\begin{bmatrix} Z_{22} & -Z_{12} \\ -Z_{21} & Z_{11} \end{bmatrix}}{\Delta Z} = \begin{bmatrix} Z_{22}/\Delta Z & -Z_{12}/\Delta Z \\ -Z_{21}/\Delta Z & Z_{11}/\Delta Z \end{bmatrix}$$

$$\text{where } \Delta Z = Z_{11} Z_{22} - Z_{12} Z_{21}$$

\* Z-parameters in terms of H-parameters & Vice versa

The defining eqn of Z-parameter.

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{--- (1)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{--- (2)}$$

The defining eqn for H-parameter.

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (3)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (4)}$$

compare eqn (1) & (3).

$V_2$  has to be eliminated in (3) using (4).

$$\text{from (4)} \quad V_2 = \frac{I_2 - h_{21} I_1}{h_{22}}$$

$$V_2 = \left( \frac{1}{h_{22}} \right) I_2 - \left( \frac{h_{21}}{h_{22}} \right) I_1 \quad \text{--- (5)}$$

⑤ In ③.

$$V_1 = h_{11} I_1 + h_{12} \left[ \frac{1}{h_{22}} I_2 - \frac{h_{21}}{h_{22}} I_1 \right]$$

$$= h_{11} I_1 + \frac{h_{12}}{h_{22}} I_2 - \frac{h_{21} h_{12}}{h_{22}} I_1$$

$$= I_1 \left[ \frac{h_{11} - h_{21} h_{12}}{h_{22}} \right] + \frac{h_{12}}{h_{22}} I_2$$

$$= I_1 \left[ \frac{h_{22} h_{11} - h_{21} h_{12}}{h_{22}} \right] + \frac{h_{12}}{h_{22}} I_2$$

$$V_1 = I_1 \left[ \frac{\Delta h}{h_{22}} \right] + \frac{h_{12}}{h_{22}} I_2 \quad \text{--- (6)}$$

Compare ⑤ & ②.

$$Z_{11} = \frac{\Delta h}{h_{22}}, \quad Z_{12} = \frac{h_{12}}{h_{22}}, \quad Z_{21} = -\frac{h_{21}}{h_{22}}, \quad Z_{22} = \frac{1}{h_{22}}$$

In matrix form

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta h}{h_{22}} & \frac{h_{12}}{h_{22}} \\ -\frac{h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$$

\* H-parameter in Z-parameter

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ -\frac{Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$$

\* Z-parameters in terms of T-parameters & vice versa

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{--- (1)} \quad V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{--- (2)}$$

$$V_1 = A V_2 - B I_2 \quad \text{--- (3)} \quad V_2 = C V_1 - D I_1 \quad \text{--- (4)}$$

Comparing (1) & (3)

$V_2$  has to be eliminated in (3) using (4).

$$(4) \Rightarrow V_2 = \frac{I_1}{C} - \frac{D}{C} I_2 \quad \text{--- (5)}$$

Substitute (5) in (3)

$$V_1 = A \left[ \frac{I_1}{C} - \frac{D}{C} I_2 \right] - B I_2$$

$$= \frac{A I_1}{C} + \frac{A D}{C} I_2 - B I_2$$

$$= \frac{A}{C} I_1 + I_2 \left[ \frac{A D}{C} - B \right]$$

$$= \frac{A}{C} I_1 + I_2 \left[ \frac{A D - B C}{C} \right]$$

$$V_1 = I_2 \left[ \frac{\Delta T}{C} \right] + \frac{A}{C} I_1 \quad \text{--- (6)}$$

compare (6) & (1)

$$Z_{11} = \frac{A}{C}, \quad Z_{12} = \frac{\Delta T}{C}$$

compare (5) & (2)

$$Z_{21} = 1/C, \quad Z_{22} = D/C$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{A}{C} & \frac{\Delta T}{C} \\ \frac{1}{C} & D/C \end{bmatrix}$$

11/4 T-parameter in terms of Z-parameter.

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$$

\* Y-parameter in terms of H-parameter & vice versa:

Defining eqn of Y-parameters.

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \text{ --- (1) , } I_2 = Y_{21} V_1 + Y_{22} V_2 \text{ --- (2)}$$

Defining eqn of H-parameter.

$$V_1 = h_{11} I_1 + h_{12} V_2 \text{ --- (3) , } I_2 = h_{21} I_1 + h_{22} V_2 \text{ --- (4)}$$

Compare (4) & (3). All variables are present in (3) hence rewrite eqn in terms of (1).

$$(3) = I_1 = \frac{V_1}{h_{11}} - \frac{h_{21}}{h_{11}} V_2 \text{ --- (5)}$$

from (5) & (1)  $Y_{11} = \frac{1}{h_{11}}$  &  $Y_{12} = -\frac{h_{12}}{h_{11}}$

Substitute (5) in (4).

$$I_2 = h_{21} \left( \frac{V_1}{h_{11}} - \frac{h_{12}}{h_{11}} V_2 \right) + h_{22} V_2$$

$$= \frac{h_{21}}{h_{11}} V_1 - \frac{h_{21} h_{12}}{h_{11}} V_2 + h_{22} V_2$$

$$= \frac{h_{21}}{h_{11}} V_1 - \left( \frac{h_{21} h_{12}}{h_{11}} + h_{22} \right) V_2$$

$$= \frac{h_{21}}{h_{11}} V_1 + \left( \frac{h_{21} h_{12} + h_{22} h_{11}}{h_{11}} \right) V_2$$

$$I_2 = \frac{h_{21}}{h_{11}} V_1 + \frac{\Delta h}{h_{11}} V_2 \text{ --- (6)}$$

Compare (6) & (2).

$$Y_{21} = \frac{h_{21}}{h_{11}} \text{ & } Y_{22} = \frac{\Delta h}{h_{11}}$$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{h_{11}} & -\frac{h_{12}}{h_{11}} \\ \frac{h_{21}}{h_{11}} & \frac{\Delta h}{h_{11}} \end{bmatrix}$$

Similarly H-parameters in terms of Y-parameters

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{Y_{11}} & -\frac{Y_{12}}{Y_{11}} \\ \frac{Y_{21}}{Y_{11}} & \frac{\Delta Y}{Y_{11}} \end{bmatrix}$$



## \* Y-parameters in terms of T-parameters (ABCD parameter)

The defining equations of Y-parameters are

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \text{ --- (1)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \text{ --- (2)}$$

The defining equations of ABCD parameters,

$$V_1 = A V_2 - B I_2 \text{ --- (3)}$$

$$I_1 = C V_2 - D I_2 \text{ --- (4)}$$

from eqn (4)  $D I_2 = C V_2 - I_1$

$$I_2 = \frac{C}{D} V_2 - \frac{I_1}{D} \text{ --- (5)}$$

Substitute (5) in (3).

$$V_1 = A V_2 - B \left[ \frac{C}{D} V_2 - \frac{I_1}{D} \right]$$

$$V_1 = A V_2 - \frac{BC}{D} V_2 + \frac{B}{D} I_1$$

$$V_1 = V_2 \left[ A - \frac{BC}{D} \right] + \frac{B}{D} I_1$$

$$V_1 = V_2 \left[ \frac{AD - BC}{D} \right] + \frac{B}{D} I_1$$

$$V_1 = \frac{\Delta T}{D} V_2 + \frac{B}{D} I_1 \quad \left( \times \frac{D}{B} \right)$$

$$\frac{D}{B} V_1 = \frac{D}{B} \times \frac{\Delta T}{D} V_2 + \frac{D}{B} \times \frac{B}{D} I_1$$

$$\frac{D}{B} V_1 = \frac{\Delta T}{B} V_2 + I_1$$

$$I_1 = \frac{D}{B} V_1 - \frac{\Delta T}{B} V_2 \text{ --- (6)}$$

compare (6) & (1).

$$Y_{11} = \frac{D}{B}, \quad Y_{12} = -\frac{\Delta T}{B}$$

Rewrite eqn (3) in the form of eqn (2).

$$B I_2 = A V_2 - V_1$$

$$I_2 = \frac{A}{B} V_2 - \frac{V_1}{B}$$

$$I_2 = -\frac{1}{B} V_1 + \frac{A}{B} V_2 \text{ --- (7)}$$

$$I_2 = -\frac{1}{B} V_1 + \frac{A}{B} V_2 \quad \text{--- (7)}$$

compare eqn (7) & (2).

$$Y_{21} = -\frac{1}{B}, \quad Y_{22} = \frac{A}{B}$$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{D}{B} & -\frac{\Delta T}{B} \\ -\frac{1}{B} & \frac{A}{B} \end{bmatrix}$$

\* T-parameters in terms of Y-parameters.

The defining eqn of T-parameters

$$V_1 = AV_2 - BI_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

The defining eqn of Y-parameters

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \text{--- (3)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \text{--- (4)}$$

compare (1) & (4).

$$(4) \Rightarrow I_2 = Y_{21} V_1 + Y_{22} V_2$$

$$Y_{21} V_1 = I_2 - Y_{22} V_2$$

$$V_1 = \frac{I_2}{Y_{21}} - \frac{Y_{22}}{Y_{21}} V_2$$

$$V_1 = -\frac{Y_{22}}{Y_{21}} V_2 + \frac{I_2}{Y_{21}} \quad \text{--- (5)}$$

compare (5) in (3)

$$I_1 = Y_{11} \left[ -\frac{Y_{22}}{Y_{21}} V_2 + \frac{I_2}{Y_{21}} \right] + Y_{12} V_2$$

$$I_1 = -\frac{Y_{11} Y_{22}}{Y_{21}} V_2 + \frac{Y_{11}}{Y_{21}} I_2 + Y_{12} V_2$$

$$I_1 = V_2 \left[ Y_{12} - \frac{Y_{11} Y_{22}}{Y_{21}} \right] + \frac{Y_{11}}{Y_{21}} I_2$$

$$I_1 = V_2 \left[ -\frac{\Delta Y}{Y_{21}} \right] + \frac{Y_{11}}{Y_{21}} I_2$$

$$I_1 = \frac{-\Delta Y}{Y_{21}} V_2 + \frac{Y_{11}}{Y_{21}} I_2 \quad \text{--- (6)}$$

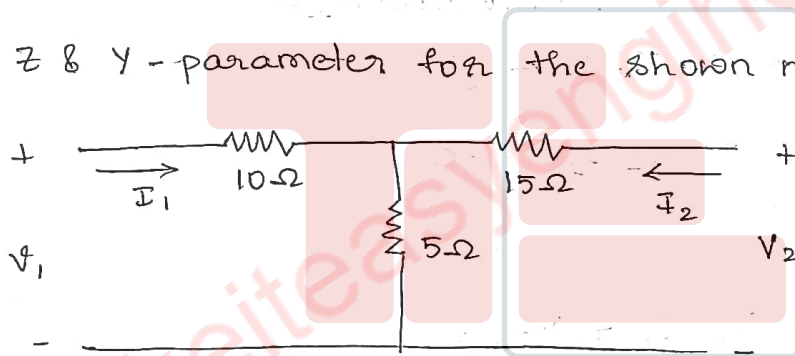
compare (6) & (2).

$$C = -\frac{\Delta Y}{Y_{21}} \quad D = \frac{Y_{11}}{Y_{21}}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} -\frac{Y_{22}}{Y_{21}} & -\frac{1}{Y_{21}} \\ -\frac{\Delta Y}{Y_{21}} & -\frac{Y_{11}}{Y_{21}} \end{bmatrix}$$

### Problems

(1) Find Z & Y-parameter for the shown network.



Z-parameter.

The defining eqn of Z-parameters are

$$V_1 = Z_{11} I_1 + Z_{12} I_2, \quad V_2 = Z_{21} I_1 + Z_{22} I_2$$

To find  $Z_{11}$  &  $Z_{21}$  make  $I_2 = 0$ .

$$Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0} \quad Z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0}$$

Apply KVL to i/p port

$$V_1 = 10 I_1 + 5 I_1$$

$$V_1 = 15 I_1$$

$$\frac{V_1}{I_1} = 15 \quad \therefore Z_{11} = 15 \Omega$$

Apply KVL to o/p port

$$V_2 = 5I_1$$

$$\frac{V_2}{I_1} = 5 \Omega, \quad Z_{21} = 5 \Omega.$$

To find  $Z_{12}$  &  $Z_{22}$  make  $I_1 = 0$ .

$$Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0}, \quad Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}.$$

Apply KVL to i/p port.

$$V_1 = 5I_2$$

$$\frac{V_1}{I_2} = 5 \quad \therefore Z_{12} = 5 \Omega.$$

Apply KVL to o/p port

$$V_2 = 5I_2 + 15I_2$$

$$V_2 = 20I_2$$

$$\frac{V_2}{I_2} = 20 \Omega \quad \therefore Z_{22} = 20 \Omega.$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 15 & 5 \\ 5 & 20 \end{bmatrix}$$

$$[Y] = [Z]^{-1}$$

$$[Y] = \frac{\text{adj}}{\Delta Z}$$

Y - parameters

The defining eqn for Y-parameters

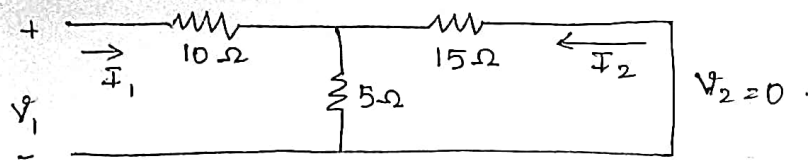
$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

To find  $Y_{11}$  &  $Y_{21}$  make  $V_2 = 0$  (i.e. short ckt the o/p port)

$$Y_{11} = \frac{I_1}{V_1}, \quad Y_{21} = \frac{I_2}{V_1}$$





Apply KVL to i/p loop.

$$V_1 = 10 I_1 + 5 (I_1 + I_2)$$

$$V_1 = 15 I_1 + 5 I_2 \quad \text{--- (1)}$$

from ckt  $I_2 = \frac{5}{5+15} \times -I_1$

$$I_2 = -\frac{5}{20} I_1$$

$$I_2 = -0.25 I_1 \quad \text{--- (2)}$$

from eqn (2)  $I_1 = -\frac{I_2}{0.25} = -4 I_2$

Substitute  $I_2$  in (1).

$$V_1 = 15 I_1 - (0.25 I_1) 5$$

$$V_1 = 13.75 I_1$$

$$\frac{V_1}{I_1} = 13.75$$

$$\frac{I_1}{V_1} = \frac{1}{13.75}$$

$$Y_{11} = 0.072 \text{ S}$$

Substitute  $I_1$  in (1).

$$V_1 = 15 (-4 I_2) + 5 I_2$$

$$V_1 = -55 I_2$$

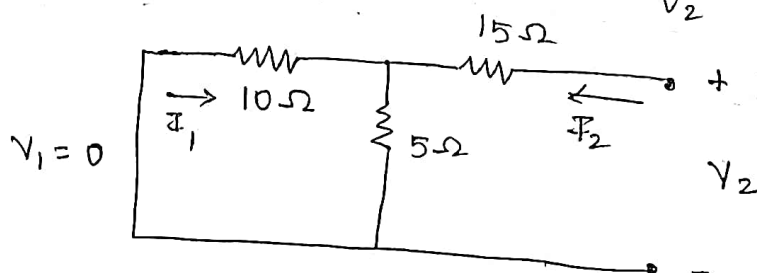
$$\frac{V_1}{I_2} = -55$$

$$\frac{I_2}{V_1} = -0.018 \text{ S}$$

$$Y_{21} = -0.018 \text{ S}$$

To find  $Y_{12}$  &  $Y_{22}$  make  $V_1 = 0$  (short ckt i/p)

$$Y_{12} = \frac{I_1}{V_2} \quad \& \quad Y_{22} = \frac{I_2}{V_2}$$



Apply KVL to o/p port

$$V_2 = 15 I_2 + 5 (I_2 + I_1)$$

$$V_2 = 20 I_2 + 5 I_1 \quad \text{--- (4)}$$

$$I_2 = 0.0545 V_2 \quad \text{--- (5)}$$

$$Z_{22} \text{ o/p impedance} = 15 + 10 \parallel 5 = 18.33 \Omega$$

$$\therefore \text{o/p admittance } Y_{22} = \frac{1}{Z_{22}} = 0.0545$$

$$\frac{I_2}{V_2} = 0.0545 \quad \text{--- (3)}$$

Substitute  $I_2$  in (4).

$$V_2 = 20 (0.0545 V_2) + 5 I_1$$

$$-0.09 V_2 = 5 I_1$$

$$\frac{I_1}{V_2} = -\frac{0.09}{5}$$

$$Y_{12} = -0.018$$

$$\therefore \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} 0.072 & -0.018 \\ -0.018 & 0.0545 \end{bmatrix}$$

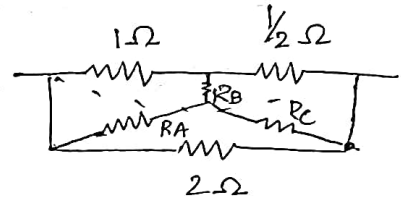
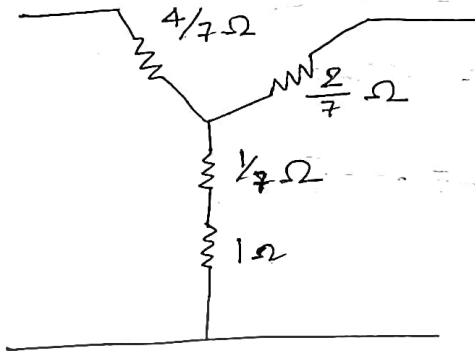
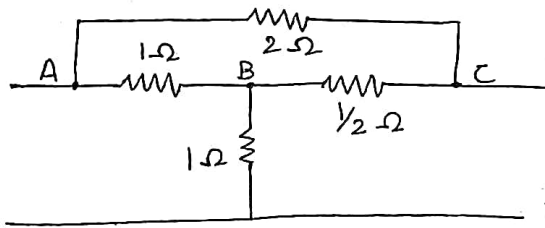
cross check:  $[Y] = [Z]^{-1}$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \frac{\text{Adj } Z}{\Delta Z} = \frac{\begin{bmatrix} Z_{22} & -Z_{12} \\ -Z_{21} & Z_{11} \end{bmatrix}}{\Delta Z}$$

$$\text{where } \Delta Z = Z_{11} Z_{22} - Z_{12} Z_{21}$$

Note :- To determine  $Y$ -parameters convert the n/w in the  $\pi$ -n/w & while finding  $Z$ -parameters convert n/w into  $T$ -n/w. This conversion simplifies the procedure to find  $Z$  &  $Y$ -parameters.

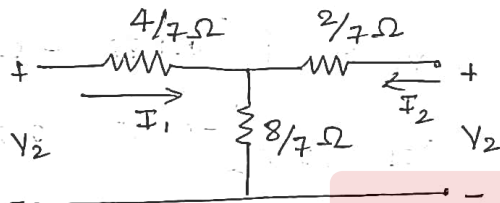
(1) Determine h parameter for n/w shown.



$$R_A = \frac{1 \times 2}{1 + 2 + 1/2} = \frac{4}{7}$$

$$R_B = \frac{1 \times 1/2}{1 + 2 + 1/2} = \frac{1}{7}$$

$$R_C = \frac{2 \times 1/2}{1 + 2 + 1/2} = \frac{2}{7}$$



To find h-parameters are

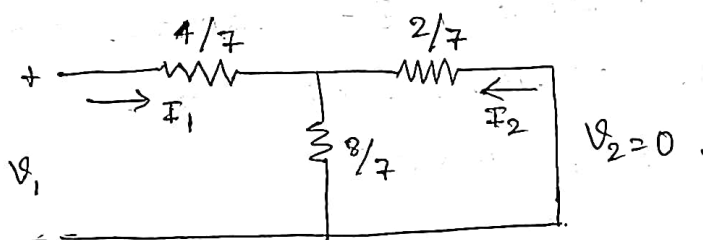
Defining eqn for h parameters are

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

To find  $h_{11}$  &  $h_{21}$  make  $V_2 = 0$  (short circuit the o/p port)

$$h_{11} = \frac{V_1}{I_1}, \quad h_{21} = \frac{I_2}{I_1}$$



$h_{11}$  = i/p impedance.

$$h_{11} = \frac{\frac{2}{7} \cdot \frac{8}{7}}{\frac{2}{7} + \frac{8}{7}} + \frac{4}{7} = \frac{4}{5} = 0.8$$

$$h_{21} = ?$$

Apply KVL to o/p loop.

$$\frac{2}{7} I_2 + \frac{8}{7} (I_1 + I_2) = 0.$$

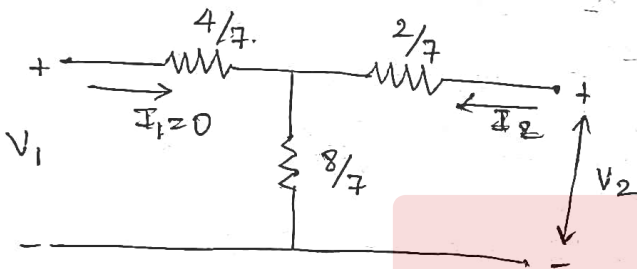
$$\frac{8}{7} I_1 + \frac{10}{7} I_2 = 0.$$

$$\frac{8}{7} I_1 = -\frac{10}{7} I_2.$$

$$\therefore \frac{I_2}{I_1} = -0.8.$$

$$\therefore h_{21} = -0.8.$$

To find  $h_{12}$  &  $h_{22}$  make  $I_1 = 0$ .



$$h_{12} = \frac{V_1}{V_2}, \quad h_{22} = \frac{I_2}{V_2}$$

Apply KVL to o/p loop.

$$V_2 = \frac{2}{7} I_2 + \frac{8}{7} I_2$$

$$V_2 = 1.428 I_2$$

$$\therefore \frac{I_2}{V_2} = \frac{1}{1.428} = 0.7 = h_{22}$$

Apply KVL to i/p loop

$$V_1 = \frac{8}{7} I_2$$

$$V_1 = \frac{8}{7} \left( \frac{V_2}{1.428} \right)$$

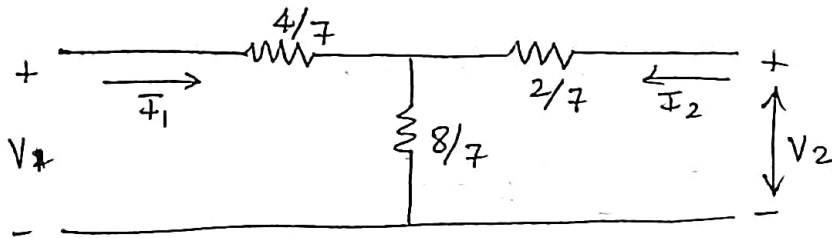
$$V_1 = 0.8 V_2$$

$$h_{12} = \frac{V_1}{V_2} = 0.8$$

$$h = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$$

$$h = \begin{bmatrix} 0.8 & 0.8 \\ -0.8 & 0.7 \end{bmatrix}$$

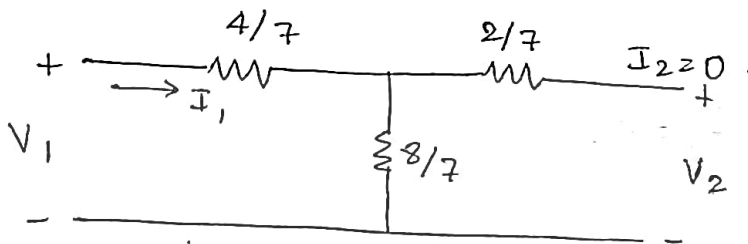
v) Determine the T-parameter for the n/w shown.



Defining eqn<sup>s</sup> :  $V_1 = AV_2 - BI_2$

$I_1 = CV_2 - DI_2$

To find A & C make  $I_2 = 0$ .



$A = \frac{V_1}{V_2}$

$C = \frac{I_1}{V_2}$

Apply KVL to o/p loop.

$V_2 = I_1 \cdot \frac{8}{7}$

$C = \frac{I_1}{V_2} = \frac{7}{8} = 0.875 \quad \text{--- (1)}$

Apply KVL to i/p loop.

$V_1 = \frac{4}{7} I_1 + \frac{8}{7} I_1$

$V_1 = \frac{12}{7} I_1 \quad \text{--- (2)}$

from (1)  $I_1 = 0.875 V_2$

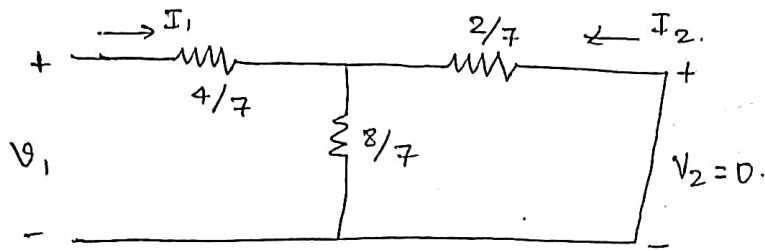
$V_1 = \frac{12}{7} (0.875 V_2)$

$A = \frac{V_1}{V_2} = 1.52$



To find B & D make  $V_2 = 0$  (Short circuit o/p port).

$$B = -\frac{V_1}{I_2}, \quad D = -\frac{I_1}{I_2}$$



Apply KVL to o/p loop.

$$\frac{2}{7} I_2 + \frac{8}{7} (I_1 + I_2) = 0.$$

$$\frac{2}{7} I_2 + \frac{8}{7} I_1 + \frac{8}{7} I_2 = 0.$$

$$\frac{8}{7} I_1 = -\frac{10}{7} I_2$$

$$\frac{I_1}{I_2} = -1.25 = -\frac{5}{4}$$

$$D = -\left(-\frac{5}{4}\right) = \frac{5}{4}$$

Apply KVL to i/p loop.

$$V_1 = I_1 \frac{4}{7} + \frac{8}{7} (I_1 + I_2)$$

$$V_1 = \frac{12}{7} I_1 + \frac{8}{7} I_2$$

Substitute  $I_1$  in (2).

$$V_1 = \frac{12}{7} (-1.25 I_2) + \frac{8}{7} I_2$$

$$V_1 = -I_2$$

$$\frac{V_1}{I_2} = -1$$

$$B = -(-1) = 1$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1.49 & 1 \\ 0.87 & 1.25 \end{bmatrix}$$